

Lab 3: ACF, PACF, and white-noise tests

Sample autocorrelations on simulated and real series

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Goals

- Compute and visualise the sample ACF and PACF.
- Verify Bartlett's bands on a simulated white noise.
- Check the theoretical ACF/PACF signatures of AR(1) and MA(1) against simulations.
- Apply ACF/PACF to the Atlanta series and run the Ljung–Box test to assess white-noise residuals.

Setup

- Python 3 with `numpy`, `statsmodels`, `matplotlib`.
- `numpy.random.seed(42)`.
- Dataset: `data/AvTempAtlanta.txt` (1656 monthly obs).

Exercise 1 – White noise simulation and ACF

- (1) Simulate a Gaussian white noise of length $T = 500$: $\varepsilon_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$.
- (2) Plot ε_t vs. t . Plot the sample ACF up to lag 30 with the $\pm 1.96/\sqrt{T}$ Bartlett bands.
- (3) Count the proportion of lags $h \in \{1, \dots, 30\}$ where $|\hat{\rho}(h)| > 1.96/\sqrt{T}$. Under H_0 this should be close to 5%. Comment.

Exercise 2 – AR(1) simulation and theoretical ACF

- (1) For $\phi \in \{0.3, 0.7, -0.7, 0.95\}$, simulate $X_t = \phi X_{t-1} + \varepsilon_t$ of length $T = 500$ with $\varepsilon_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$, $X_0 = 0$.
- (2) For each ϕ , plot the empirical ACF up to lag 20 and overlay $\rho_{\text{theo}}(h) = \phi^h$ as red points.
- (3) Plot the empirical PACF for $\phi = 0.7$. Is it cut off at lag 1, as the theory predicts?
- (4) For $\phi = 0.95$, the series is *nearly* non-stationary. What does the empirical ACF look like? How does it differ from the theoretical 0.95^h on a finite sample of size 500?

Exercise 3 – MA(1) simulation

- (1) For $\theta \in \{0.5, -0.5, 0.9\}$, simulate $X_t = \varepsilon_t + \theta \varepsilon_{t-1}$ of length $T = 500$.
- (2) Plot the empirical ACF and verify that:
 - $\hat{\rho}(1) \approx \theta/(1 + \theta^2)$;
 - $\hat{\rho}(h) \approx 0$ for $h \geq 2$.

(3) Plot the empirical PACF: is it decaying as expected?

Exercise 4 – ACF / PACF on Atlanta temperatures

Load `data/AvTempAtlanta.txt` and stack to a monthly series ($n = 1656$).

- (1) Plot the raw ACF up to lag 60. What do you see? Justify the strong spikes at lags 12, 24, 36.
- (2) Decompose the series (Chapter 2, additive workflow with 2×12 MA + seasonal averaging) and compute the residual $\hat{\varepsilon}_t$.
- (3) Plot the ACF and PACF of $\hat{\varepsilon}_t$. Are the spikes at lags 12, 24, 36 still present? What does this say about the quality of the decomposition?

Exercise 5 – Ljung–Box test

- (1) Use `statsmodels.stats.diagnostic.acorr_ljungbox` to test, on the residual $\hat{\varepsilon}_t$ of Exercise 4: H_0 : residuals are white noise, vs. H_1 : residuals have autocorrelation, at lags $m \in \{12, 24, 36\}$ and significance level $\alpha = 5\%$.
- (2) Report the test statistic, p -value, and conclusion at each lag. What does this tell you about the residual?
- (3) [bonus] Apply the same test to the simulated white noise of Exercise 1 — you should *not* reject H_0 .

Warning

The functions `statsmodels.graphics.tsaplots.plot_acf` and `plot_pacf` draw the Bartlett bands automatically.